

P3 Differentiation

Differentiation of standard functions

Functions (y)	Derivatives $\left(\frac{dy}{dx}\right)$
constant, k	0
x^n	nx^{n-1}
e^x	e^x
a^x ive constant	$a^x \ln a$ $\frac{d}{dx}(3^x) = 3^x \ln 3$
$\ln x$	$\frac{1}{x}$ $\frac{d}{dx}[\ln(x^2 + 3x + 5)] = \frac{1}{x^2 + 3x + 5} (2x + 3)$
$\sin x$	$\cos x$ $\frac{d}{dx}(\sin 2x) = (\cos 2x)(2) = 2 \cos 2x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$ $\frac{d}{dx}(\tan 3x) = (\sec^2 3x)(3) = 3 \sec^2 3x$
$\tan^{-1} x$	$\frac{1}{1+x^2}$ $\frac{d}{dx}(\tan^{-1}(3x)) = \frac{1}{1+(3x)^2} (3) = \frac{3}{1+9x^2}$

Differentiation of common functions

Functions (y)	Derivatives $\left(\frac{dy}{dx}\right)$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{1}{x}$	$-\frac{1}{x^2}$
e^{kx}	ke^{kx}
$\ln(ax + b)$	$\frac{a}{ax + b}$
$\sin kx$	$k \cos kx$
$\cos kx$	$-k \sin kx$
$\tan kx$	$k \sec^2 kx$

Differentiating Hyperbolic Functions

1. The standard results are:

(a) $\frac{d}{dx}(\cosh x) = \sinh x$	$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
(b) $\frac{d}{dx}(\sinh x) = \cosh x$	$\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$
(c) $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$	$\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$

Recall	
$\frac{d}{dx}(\cos x) = -\sin x$	$\frac{d}{dx}(\sec x) = \sec x \tan x$
$\frac{d}{dx}(\sin x) = \cos x$	$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

$$(a) \frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$$

$$(b) \frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2 + 1}}$$

$$(c) \frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1 - x^2}$$

